

MANTHAN PENDKE

AI/ML Engineer | Applied Machine Learning & LLM Systems

+91 8482916576 | mpendke9@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Entry-level AI/ML Engineer and published IEEE researcher with internship experience building LLM-based systems, including Retrieval-Augmented Generation (RAG) pipelines, LLaMA-based models, and machine learning models on imbalanced data using SMOTE. Built a production-style AI support assistant with structured LLM output, knowledge-base grounding, automated evaluation and CI, and delivered ML-powered dashboards with React.js and REST APIs. AWS Certified Cloud Practitioner seeking to apply model fine-tuning, MLOps and full-stack skills to ship reliable AI products.

TECHNICAL SKILLS

AI/ML & LLMs: RAG, LLaMA, LangChain, FAISS, Vector Databases, TF-IDF Retrieval, Groq API, Prompt Engineering, Structured LLM Output, LLM Evaluation, NLP, Computer Vision, OpenCV, TensorFlow, PyTorch, SMOTE

ML Workflow: Model Fine-Tuning, Classification, Root-Cause Analysis, Recommendation Systems, MLOps (Deployment & Monitoring), Automated Evaluation, Regression Testing (Pytest)

Languages: Python, Core Java, JavaScript, C++

Backend & Frontend: FastAPI, Pydantic, REST APIs, Node.js, Express.js, React.js, Streamlit, HTML5, CSS3

Databases: SQL Server, MySQL, MongoDB

Cloud & DevOps: AWS, Git, GitHub, GitHub Actions (CI/CD)

PROFESSIONAL EXPERIENCE

Developer Intern | Endava Solutions

January 2026 – May 2026

- Developed a personalized recommendation system using content-based filtering, collaborative filtering (user-based and item-based) and matrix factorization to generate tailored recommendations.
- Designed a relational database schema in SQL Server using ER modeling and normalization, structuring data for downstream ML and feature-store use.
- Built backend modules in Java using OOP principles for data processing and business logic implementation.
- Performed functional, database and UI testing to validate system reliability and data integrity.

AI Intern | Expleo

July 2025 – August 2025

- Developed a predictive maintenance system for CNC machines using machine learning and LLaMA-based LLMs to forecast alarms and minimize downtime.
- Implemented Retrieval-Augmented Generation (RAG) and SMOTE to enhance alarm diagnostics and improve model performance on imbalanced datasets.
- Built and fine-tuned ML models for alarm classification and root-cause analysis using historical sensor and operational data.
- Designed a React.js dashboard for real-time alarm visualization, integrating predictive analytics directly into the CNC machine workflow.

PROJECTS

AI Support & TAM Assistant | Python, FastAPI, Groq (LLaMA 3.3 70B), RAG, Pydantic, Streamlit, GitHub Actions

- Built a RAG-based ticket triage pipeline (TF-IDF retrieval + Groq LLM) over 500 synthetic tickets and a Markdown knowledge base, returning validated JSON with product, category, P1–P4 urgency, known-issue status, responder team and draft reply.
- Grounded LLM outputs in retrieved knowledge-base evidence using Pydantic validation and grounding checks to prevent fabricated claims.
- Engineered a deterministic account-health engine (90-day ticket trends, seat utilization, usage trend, NPS, renewal timing) across 50 accounts; the LLM generates the executive TAM summary but cannot override the score.
- Implemented data-quality checks that flag inconsistencies between structured account fields and escalation notes.
- Automated a 12-case evaluation harness (including adversarial cases, 100% pass rate) and an 18-test Pytest suite in GitHub Actions CI, with versioned prompts.
- Deployed a FastAPI backend with a streaming triage endpoint and a Streamlit UI for non-technical users.

Disaster Information Aggregation Software | Deep Learning, Computer Vision, OCR, NLP

- Developed a multimodal disaster classification system using deep learning to identify disaster types and assess severity from images and unstructured data.
- Implemented computer vision pipelines (image preprocessing, augmentation, OCR, object detection) to improve model performance.
- Integrated visual, textual and geospatial data into an end-to-end real-time disaster intelligence and alerting system.
- Published results as an IEEE research paper covering the multimodal NLP, computer vision and ML architecture.

College Prediction System | Machine Learning, MLOps

- Developed a machine learning-based admission prediction system to recommend eligible colleges, branches and admission rounds from student rank and category.
- Designed custom prediction algorithms using historical admission data for personalized recommendations.
- Automated data preprocessing, model training and prediction workflows, applying MLOps practices for deployment and monitoring.

Location-Optimized Web System for Jobs | React.js, Node.js, MongoDB

- Built a recruitment platform with location-based search and proximity-driven recommendation logic, with secure authentication and real-time application tracking.

EDUCATION

B.Tech, Electronics & Telecommunication Engineering

November 2022 – June 2026

CGPA: 8.89

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services

May 2026

Introduction to Large Language Models (LLMs) | Udemy

April 2025

PUBLICATION

- **IEEE Research Paper:** Disaster Information Aggregation using Multimodal Artificial Intelligence, integrating NLP, Computer Vision and Machine Learning.